

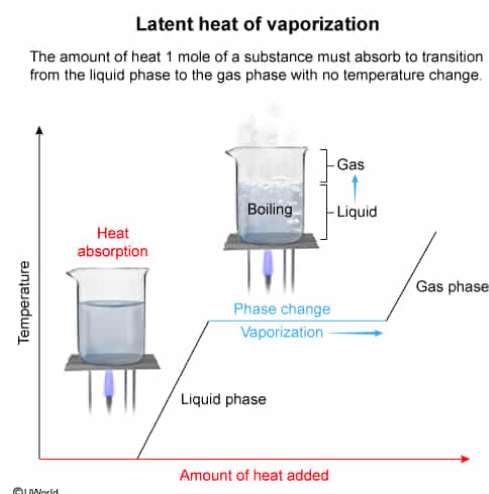
The latent heat of vaporization describes:

- ☐ A. the amount of heat a substance must release to transition from the liquid to the gas phase. [11%]
- ✓ ☒ B. the amount of heat a substance must absorb to transition from the liquid to the gas phase. [85%]
- ☐ C. the amount of heat a substance must release to transition from the solid to the gas phase. [1%]
- ☐ D. the amount of heat a substance must absorb to transition from the solid to the gas phase. [1%]

Correct

85% answered correctly

Explanation:



Latent heat is the amount of heat 1 mole of a substance must absorb (or release) to produce a **change in phase** (eg, solid to liquid, gas to liquid) while maintaining a **constant temperature**.

When a substance in the liquid phase is at its boiling point, the latent heat energy that goes into the system is used to break the attractive **intermolecular forces** between the liquid molecules rather than causing an increase in temperature. Consequently, the energized liquid molecules can escape the surface of the liquid and enter the gas (ie, vapor) phase.

Therefore, the amount of **heat absorption** needed to convert 1 mole of a substance in the **liquid phase to the gas phase** is described by the **latent heat of vaporization** ($\Delta H_{\text{vaporization}}$).

(Choice A) A substance in the liquid phase must *absorb* heat to vaporize (ie, transition from the *liquid* phase to the *gas* phase).

(Choices C and D) The amount of heat *absorption* needed to convert 1 mole of a substance in the solid phase to the gas phase is described by the latent heat of *sublimation*.

Educational objective:

The latent heat of vaporization of a substance describes the heat required to convert 1 mole of that substance from the liquid phase to the gas phase.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403189

System:
Thermodynamics, Kinetics & Gas Laws